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Article

Journal of Career Development
2022, Vol. 49(4) 862–874
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DOI: 10.1177/08948453211000130
journals.sagepub.com/home/jcd
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Dysfunctional Career Thoughts and the Sophomore Slump Among Students With Learning Disabilities

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Abstract

The present study sought to underline the need for expanded transitioning college programming to the 2nd year by examining dysfunctional career thoughts among college students with learning disabilities (LDs). Multivariate analysis of variance (MANOVA) was used to examine the mean differences between 93 college sophomores and seniors with LD on the three subscales of the Career Thoughts Inventory (CTI). Findings from the MANOVA showed significant mean differences among groups. Follow-up analysis found that sophomores showed significantly higher dysfunctional career thoughts than seniors. Seniors displayed substantially lower mean scores on the CTI subscales than sophomores. Findings support the need to extend transition programming that includes addressing students' dysfunctional career thoughts beyond the 1st year to help ease college transition difficulties and thus promote persistence and retention of students with LD. Research and practice implications are presented for career scholars and practitioners working with college students with LD.

Keywords

dysfunctional career thoughts, college students with learning disabilities, sophomore slump, college persistence

Students with learning disabilities (LD) make up the largest group of college freshmen with disabilities (Goegan & Daniels, 2019). Though the number of students with LD at colleges has steadily been on the rise (National Center for Education Statistics, 2019; Snyder et al., 2016), retention takes a sharp decline after the 1st year (Gregg, 2007; Schaller, 2005). Students with disabilities (SWD) have lower retention rates than all other students (Lombardi et al., 2012), with about 25% not continuing after their

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1st year (Ketevan & Koch, 2012). Evidence suggests that the 2nd year can be more challenging than the 1st year (Gahagan & Hunter, 2006; Harris, 2018) due to higher academic demands, students' low self-efficacy, and career indecisiveness (Gardner, 2000; Harris, 2018). Among SWD specifically lower retention rates are associated with additional obstacles they encounter in maintaining persistence (e.g., limited academic accommodations, career-related transition programming, and support; Cortiella & Horowitz, 2014; Ketevan & Koch, 2012). Hence, the present study sought to shed light on the need to improve and expand transition-related programming (Roessler et al., 2009) by examining negative career thoughts among sophomore and senior college students with LD.

The Sophomore Slump

College administrators are charged with providing newly arriving students with retention strategies to overcome the cognitive, social, and transitional challenges related to the college experience (Newman & Madaus, 2015; Williams & Butler, 2010). However, transitional programming and interventions are generally "front-loaded" and fail to carry into the 2nd year (Gahagan & Hunter, 2006; Harris, 2018). College administrators tend to believe that if they succeeded in retaining students through the 1st year, these efforts should be repeated in subsequent years with each 1st-year class (Williams & Butler, 2010). Thus, the 2nd year is often a time when students who are likely still acclimating to the college environment receive less support (Harris, 2018; Schroeder & Gardner, 2003; Wang & Kennedy-Phillips, 2013).

Researchers report evidence of what is described as the "sophomore slump," a phenomenon that gives credence to the transitional needs of 2nd-year college students, defined as the feeling of being discontented with the college experience (Wang & Kennedy-Phillips, 2013, p. 541). During this period, students embrace their autonomy but struggle to establish their own unique identity and develop a sense of purpose. The sophomore slump is associated with several negative effects including confusion, depression, and anxiety (Harris, 2018; Lemons & Richmond, 1987); inertia, disorganization (Tobolowsky & Cox, 2007), career indecisiveness, and lack of clarity about what it takes to convert academic goals into career choice (Gardner, 2000; Harris, 2018).

Sophomore students, for example, are likely to exhibit lower self-efficacy, increased self-doubt, and less resilience as compared to their peers, resulting in dysfunctional thoughts about the college experience and their postcollege future (Harris, 2018). Dysfunctional or negative thinking can hinder students' college success and ultimately contribute to the decision to withdraw from college (Thompson-Ebanks, 2014). Thus, the sophomore slump is considered a significant hurdle to students' persistence to graduation without intervention. As such, sophomore students need guidance, career intervention, and more support in college (Hunter et al., 2010).

College Students With LD

Students' time in college constitutes a period of tremendous change accompanied by self-discoveries with endless possibilities (Hunter et al., 2010; Schaller, 2005). This period coincides with emerging adulthood (18–25 years of age), considered one of the most stressful developmental transitions (Arnett, 2015; Ng & Feldman, 2007). During this time, young adults undergo physiological and psychological changes that influence their identity, self-concept, and, ultimately, career development (Dipeolu et al., 2019). This period is undoubtedly a crucial time for students' general identity development (Arnett, 2015), vocational identity progression (Osborn et al., 2007), and the exploration of career choices.

SWD face additional challenges related to growth and development in the context of multiple identities (Wood et al., 2017) as compared to students without disabilities, particularly first-generation SWD or ethnic minority SWD. Students with LD experience academic challenges coupled with the stigma associated with disability in the context of multiple identities, which may negatively influence

and undermine students' belief in themselves (Kim & Kutscher, 2020). Negative beliefs or dysfunctional career thoughts (i.e., I can never be successful in college, college is for smart people, not for me, etc.) can influence every aspect of students' functioning, including career decision making, planning, and goal setting (Sampson et al., 2004). While all students may manifest negative beliefs that impede their career decision-making process (Dipeolu et al., 2002), students with LD are particularly vulnerable to these thoughts due to their disability status (Authors, 2019; Woodcock & Vialle, 2010). The complex and unique challenges faced by SWD could provide one explanation for why these students report less satisfaction with the college experience (McGregor et al., 2016). Together, these experiences perpetuate negative dysfunctional thoughts that may impede them to persist through college and become successful later in life.

Theoretical Framework

Research within the last 20 years has used the cognitive information processing (CIP) to explain career decision making and identify dysfunctional/negative career thoughts as part of the college transition intervention program among young adults with and without LD (Dipeolu et al., 2012, Dipeolu et al., 2019; Sampson, 2017). Using CIP, Sampson et al. (1996) theorized about the effects of negative career thoughts on students transitioning to college and the world of work. The CIP model's central tenet is that the primary goal of career counseling is to help clients develop information-processing skills that will improve their ability to make independent career decisions (Sampson et al., 1996; Sampson et al., 2004).

According to the CIP model, an essential aspect of the career intervention and planning process is the acquisition of self-knowledge derived from individuals' life experiences. However, some individuals develop dysfunctional career thoughts that may impede their ability to make decisions, achieve their life goals (Sampson et al., 2004), and help to persist through college. For example, negative career thoughts may contribute to indecisiveness by preventing individuals from learning new metacognitive skills necessary for career problem solving (Carr, 2004). As previously stated, college SWD tend to show more negative attributions as compared with college students without disabilities (Reiff, 2004; Woodcock & Vialle, 2010). These negative attributions can translate into negative schemas that guide how an individual processes information, solves problems, makes decisions, and "thinks about thinking" (Peterson et al., 1991, p. 12). In essence, deeply embedded negative attributions which lead to dysfunctional thoughts can result in anxiety and career decision-making difficulty for students with LD (Dipeolu et al., 2002) and could present a challenge navigating through transitions while in college.

Using the CIP approach, students with LD can integrate three domains of functioning relevant to an understanding of (1) information processing, (2) decision making, and (3) executive processing to make effective career decisions. The information-processing dimension includes the constructs of self-knowledge, occupational knowledge, decision-making skills, and clients' metacognition (Sampson et al., 1996, 2004). The second dimension of the approach outlines the necessary skills for making career decisions, including the pyramid of information processing domains, the Communication, Analysis, Synthesis, Valuing, & Execution (CASVE) cycle, a decision-making heuristic that includes communication, analysis, synthesis, valuing, and execution stages of career intervention process (Dieringer et al., 2017). Dysfunctional career thoughts can occur in all areas of the pyramid and CASVE cycle (Sampson et al., 2004) and can be assessed using the CTI (Sampson et al., 1996). Last, the third domain emphasizes executive processing, including metacognitive skills of self-awareness, self-talk, control, and monitoring used in the career decision-making process (Sampson et al., 1996, 2004).

Dysfunctional Career Thoughts Among Students With and Without LD

Although research on college students with LDs and dysfunctional career thoughts is scant, the few existing studies generally show that students with LDs have fewer dysfunctional career thoughts than normative samples (Dipeolu et al., 2002) on the CTI Total, Decision-Making Confusion (DMC), and Commitment Anxiety (CA) Scales of instruments developed to assess for the presence of dysfunctional career thoughts. However, on the External Conflict (EC) Scale of the same measure, students with LDs exhibit higher dysfunctional career thoughts than normative samples (Dipeolu et al., 2002). A majority (75%) of students who participated in these studies are in the upper division (38% were seniors, 27% juniors; Dipeolu et al., 2002). Thus, the findings regarding dysfunctional career thoughts could be considered an overestimation of the extent to which negative career thoughts are exhibited by students since most participants are closer to completing their degree. As such, to understand the patterns of dysfunctional career thoughts among college students with LDs, a study that compares students in the lower division, in this case, the sophomore, with those in the upper division (i.e., seniors), is a crucial next step in this line of inquiry.

Squires et al. (2018) suggest programmatic offerings based on different academic standing (e.g., 1st year vs. 4th year) as a logical focus of research. This is a reasonable proposition since age is related to frequency of dysfunctional career thoughts, with younger students reporting more dysfunctional career thoughts than older students (Dipeolu et al., 2002). This line of reasoning is also consistent with the findings on age in the normative study, which consists of SWD (Sampson et al., 1996; 2004). Based on these findings, we speculated that sophomores with LD are likely to exhibit more dysfunctional career thoughts than college seniors with LD. The sophomore year could be considered a better anchor point in a study that seeks to examine the case for extending transition programming beyond the 1st year for students with LD. As such, we examine dysfunctional career thoughts among sophomores with LD as compared to seniors with LD to advance our understanding of sophomore slump, college persistence, and college transitional programming. The current study sought to add knowledge to the field by examining the following questions: (1) Do sophomore students with LD score significantly higher than senior students with LD on the three scales of the CTI related to DMC, CA, and EC? (2) Are sophomore students with LD scores different from the norms published in the CTI manual for college students in general? and (3) Do seniors with LD score different from norms published in the CTI manual for college students?

Method

Participants

The present study consisted of a total of 93 undergraduate students. The sample included 29 sophomores (31.2%) and 64 seniors (68.8%). LDs represented in the sample included written language (26.9%), reading, (32.3%), math (46.9%), other (30%), and unknown (5.4%). Participants ranged in age from 19 to 47 years ($M = 24.83$; $SD = 6.06$). The sophomore group consisted of 10 females (34.5%) and 19 males (65.5%), and the senior group consisted of 29 females (45.3%) and 35 males (54.7%). A Mann–Whitney U test was conducted to determine whether the distribution of gender in the two groups was significantly different. Results were nonsignificant, suggesting that there was no significant difference between the two groups ($U = 908.00$, $p > .05$). Ethnicity for the sophomore group included one African American (3.4%), 24 European American (82.8%), three Asian (11.8%), 0 Hispanic (0%), and one with ethnicity identified as Other (3.4%). Senior group ethnicity comprised nine African American (14.1%), 47 European American (73.4%), 0 Asian (0%), five Hispanic (7.8%), and three ethnicity identified as Other (4.7%). As with gender, a Mann–Whitney U test was conducted and found no significant differences in the distribution of ethnicity between the groups ($U = 840.00$, $p > .05$). The population studied can be considered rare or hard-to-reach, and thus the method of sampling "based on the principle of finding

out where persons are present . . . and establishing an unbiased estimator, *a posteriori* for the entire population" was used (Marpsat & Razafindratisma, 2010, p. 5). Specifically, the current study utilized a limited, purposive sample from the identified members of the larger, hard-to-reach population to estimate population parameters.

Instruments

Career Thoughts Inventory (CTI). The CTI (Sampson et al., 1996) was used to examine negative/dysfunctional career thoughts. Although the CTI is a 48-item instrument, 29 items were used for the three scales of dysfunctional career thoughts: DMC-14 items, CA-10 items, and EC-five items. Each item on the scale is designed to represent a dysfunctional career-related thought. Respondents are asked to identify their agreement with each statement, with response options ranging from 0 (*strongly disagree*) to 3 (*strongly agree*). The total score provides a global evaluation of negative/dysfunctional career thoughts.

The CTI's items and scales possess content, construct, and criterion-related validity at acceptable levels (Sampson et al., 1996). The content validity of the instrument is based on the CIP approach to career decision making. Test–retest reliability was initially measured in college and high school students across 4 weeks ranging from .74 to .82 (Sampson et al., 1996). The internal consistency for a sample of undergraduate students without LD was comparable to that of undergraduate college students with LD (Dipeolu, et al., 2002). The internal consistency for the CTI is high, with a coefficient α of .96. Dipeolu et al.,(2012) found support for the reliability of this measure with high school students with LD, with α coefficients of .95, .89, .80, and .75 for CTI total score and the DMC, CA, and EC subscales, respectively. The current study had α coefficients for the CTI ranging from .78 to .93, with coefficients of .93, .89, and .78 for the DMC, CA, and EC scores, respectively.

Procedure

The university's institution review board approved the research protocol for the study before data collection. Participants were recruited through the office for SWD at a southeastern university in the United States. A number of 460 students were contacted from which only 103 (22%) responded and were invited to an information session regarding the study. Interested students received information about the study's rationale and procedures in a separate room from the principal researcher. These procedures ensured that all aspects of the research process, including potential risks, benefits, confidentiality, and voluntary participation, were covered. Questions and concerns were answered and addressed accordingly. Participants completed the informed consent before data collection. Participants were provided with the option to withdraw from the research process at any point during the data collection process. Students who signed the consent form also completed a datasheet to collect demographic information and the CTI. Of the 93 participants, six received a reader's assistance, the accommodation offered due to the nature of disabilities, and the reading tasks involved to complete the study.

Results

A one-way multivariate analysis of variance (MANOVA) was conducted using SPSS to determine differences between college sophomores and seniors on the three dependent variables measuring negative career thinking. The dependent variables were the three scales of the CTI: DMC, CA, and EC. There were no missing data for the analysis. Significant differences were found between the seniors and sophomores on the three measures of dysfunctional career thoughts, Wilks' $\Lambda = .92$, $F(3, 89) = 2.72$, $p = .049$, partial $\eta^2 = .08$. Although the sample sizes of the two groups were different (sophomores = 29; 31.2%, seniors = 64, 68.8%), we concluded that the small number of dependent variables

Table 1. Means and Standard Deviations (*SD*s) on the Dependent Variables for the Groups.

Dysfunctional Career Thoughts Scale Means	Sophomore Level		Senior Level		College Norms [a]	
	M	*SD*	*M*	*SD*	*M*	*SD*
Decision-making confusion	11.76	7.52	7.53	6.73	10.72	7.39
Commitment anxiety	13.41	5.53	10.11	5.92	12.92	5.36
External conflict	5.03	2.77	3.55	2.88	3.32	2.15

[a]Norms from Sampson et al. (1996).

Table 2. Post Hoc Analysis of Variance Results for Career Thoughts Inventory Subscale Sophomore and Senior Students With Learning Disability.

Dysfunctional Career Thoughts Scale	*F*	*df*	Significance	Partial η^2
Decision-making confusion	7.31	(1, 91)	.008*	.07
Commitment anxiety	6.48	(1, 91)	.013*	.07
External conflict	5.45	(1, 91)	.022	.06

Note. Post hoc results using Bonferroni method with analysis of variance tested at $p < .016$.

together with a sample size of at least 20 in the smallest cell met the requirements of robustness for use identified by Mardia (1971). Table 1 contains the means and standard deviations (*SD*s) on the dependent variables for the two groups and the norms for college-aged students established in the literature.

As a follow-up to the MANOVA results, an ANOVA on each dependent variable was conducted. The univariate test on the DMC scale was significant, $F(1, 91) = 7.31$, $p = .008$, partial $\eta^2 = .07$. When comparing students with LD, seniors ($M = 7.53$, $SD = 6.73$) had significantly lower DMC than sophomores ($M = 11.76$, $SD = 7.52$). Likewise, the CA mean of the seniors ($M = 10.11$, $SD = 5.92$) was lower than the sophomores ($M = 13.41$, $SD = 5.52$). The difference between the seniors and sophomores on the CA scale was significant, $F(1, 91) = 6.48$, $p = .013$, partial $\eta^2 = .07$. The ANOVA on EC scores was also significant, $F(1, 91) = 5.45$, $p = .022$, partial $\eta^2 = .06$. On the EC scale, seniors' scores ($M = 3.55$, $SD = 2.88$) were significantly lower than the sophomores ($M = 5.03$, $SD = 2.77$; see Table 2).

Next, using the one-sample *t*-test, the college student population mean (CSPM) and means of sophomores and seniors with LD were compared. Significant differences were observed. We reported CSPM on the CTI total score and its three subscales of DMC, CA, and EC. Due to having four dependent scales (CTI-Total, DMC, CA, and EC) and an attribute variable with two levels (senior and sophomore), eight one-sample *t*-tests were conducted. To reduce the chance of making a Type I error, a Bonferroni correction was conducted, α was set at .006 and $p < .006$.

Similar to the findings in the MANOVA, these one-sample *t*-tests indicated a difference between seniors and sophomore students with LD compared to the population mean CTI score. For CTI total score, seniors with LD ($M = 37.34$, $SD = 22.47$) was almost 10 points lower than CSPM ($M = 47.01$, $SD = 20.89$), indicating that seniors with LD have significantly lower negative career thinking scores than the college student population, $t(63) = -3.44$, $p = .001$. However, the CTI total score for sophomores with LD ($M = 51.11$, $SD = 22.81$) was only slightly higher but not significantly different from the college student population, $t(28) = .97$, $p =$ ns (see Table 3).

Correspondingly, the DMC mean score for the seniors with LD ($M = 7.53$, $SD = 6.73$) was about three points lower than the CSPM-DMC score ($M = 10.72$, $SD = 7.39$). This difference suggests that seniors with LD have significantly less DMC than the CSPM, $t(63) = -3.79$, $p < .001$. However, this

Table 3. Independent Group *t*-Tests for Career Thoughts Inventory Subscale Differences.

Dysfunctional Career Thoughts Scale Means	Mean Difference	*df*	*t*
Sophomores			
CTI total	4.10	28	0.97
Decision-making confusion	1.04	28	0.74
Commitment anxiety	0.49	28	0.48
External conflict	1.71	28	3.33*
Seniors			
CTI total	−9.67	63	−3.44*
Decision-making confusion	−3.79	63	−3.79*
Commitment anxiety	−2.81	63	−3.80*
External conflict	0.23	63	0.63

Note. Post hoc results using Bonferroni correction tested at $p \leq .006$.
*$p < .01$. **$p < .001$.

difference on the DMC subscale was not observed in sophomore students with LD ($M = 11.76$, $SD = 7.52$) when compared to CSPM ($M = 10.72$, $SD = 7.39$), $t(28) = .74$, $p =$ ns.

On the CA subscale, seniors with LD had a significantly lower mean score compared to the CSPM-CA. However, this difference was not found when comparing sophomore students with LD to CSPM-CA. When analyzing the EC subscale, results revealed a significantly lower mean score for the difference between sophomore students with LD and the CSPM-EC. There was no significant difference found when senior students with LD were compared to the CSPM-EC.

Discussion

Sophomores and Seniors With LD

The present exploratory study sought to determine whether dysfunctional career thoughts would decrease as students with LD persist through college. To determine whether dysfunctional career thinking changes in college, we performed a cross-sectional analysis, comparing students' career thoughts in the sophomore year with those in the senior year on the three subscales of the CTI related to DMC, CA, and EC. Significant differences were found among the dependent variables of these three subscales. Follow-up ANOVA tests conducted on DMC suggest that seniors have significantly fewer negative career thoughts related to initiating or sustaining the decision-making process than sophomores, consistent with previous research related to college students with LD (Dipeolu et al., 2002). Related to findings on the DMC scale, decision making is one of the several skills that constitute self-determination. College students with LD exhibit higher dysfunctional career thoughts on the DMC scale comparatively in this study. Self-determination has repeatedly been shown to be an essential skill set critical to success for SWDs in higher education (Wehmeyer & Palmer, 2003). Shoring up students' decision-making skills by decreasing dysfunctional career thoughts through a transition program extended to the sophomore year (Squires et al., 2018) could therefore increase student's self-determination. This, too, will likely contribute to students with LD persistence to graduation. Thus, this finding supports extending the transition programming beyond the 1st year for students with LD.

On the CA, the follow-up ANOVA suggests that seniors have significantly fewer negative career thoughts than sophomores when it comes to committing to a specific career choice. The negative thoughts associated with the CA subscale are manifested by anxiety about the outcome of the career decision-making process, and this anxiety perpetuates career indecision. On the EC subscale, the follow-up test was significant and suggested that seniors have significantly fewer negative career thoughts about balancing the importance of their self-perceptions with the perceptions and opinions

of significant others. This difference also suggests that seniors are more willing than sophomores to assume responsibility for decision making is less affected by EC. The findings also support the conclusion that developmentally, students might age-out of specific dysfunctional thinking patterns. However, this study indicates that transition programming services tailored to students with LDs that include career intervention to address dysfunctional career thoughts related to CA and EC should be extended to the sophomore year.

The findings related to the CA and EC subscales are notable. One possible explanation for the results reflected in the CA subscale may be attributed to the inherent nature of anxiety and indecision associated with this subscale. Anxiety around career decision can be resolved by students' internal (self)reflection. In contrast, the EC subscale measures problems associated with external factors/influence (i.e., significant others) and requires involving others to resolve associated issues. This finding indicates that sophomores with LD who score higher on the EC subscale may need to learn the give-and-take of negotiating with significant others compared to senior students with LD. Thus, the sophomore year could be considered an essential developmental anchor point, one that could benefit from career-based transition interventions, with particular emphasis on EC and programmatic attention for students with LD.

Sophomores With LD and College Students Population

We examined whether there was a significant difference between sophomores and seniors on the CTI total (global dysfunctional career thoughts), DMC, CA, and EC compared to the CSPM on each scale. The analyses conducted to address this question indicated that sophomores were not significantly different from the CSPM on global dysfunctional career thoughts, DMC, and CA. However, sophomores' mean for EC was significantly higher than the CSPM-EC, while senior students' scores were similar to college students in general. This finding is consistent with existing research (Dipeolu et al., 2002, Dipeolu, 2007, Hitchings et al., 2001, 2005), which reported that regarding attribution style for career decision-making, SWD believed external factors were responsible for career-related outcomes in their lives and may benefit from transition programming that directly targets related areas.

Seniors With LD and College Students Population

By contrast, the mean for seniors is significantly lower for global dysfunctional career thoughts on the DMC and CA scales than the CSPM scales. There was no significant difference in EC between seniors with LD and the CSPM. Seniors with LD had lower EC scores. This finding suggests that on the EC subscale, seniors had a significantly lower score than the CSPM. Seniors with LD's scores were lower (less negative thoughts) than sophomores with LD's scores. As a student develops more self and occupational knowledge, they may be better equipped to engage in the career decision-making activities as noted in the CAVSE cycle. As students experience more successful decision-making experiences, they change from negative self-talk to more positive and optimistic self-talk. This conclusion is a logical proposition, given the level of maturation expected of students by the senior year of college. Currently, while we understand the need to extend programming past the 1st-year of college to increase persistence and reduce attrition, we do not have enough empirical support to precisely determine the effect of increased transition programming in the 2nd year. However, we can hypothesize about the potential efficacy of transition programs after the 1st year.

Students with LD exhibit more anxiety or nervousness about commitment to a major or career. Yet, as they persist through college, they gain more confidence, exhibit a decrease in negative thoughts about careers, learn more about careers, and identify potential career fields. However, the sophomore year remains a problematic transition period. Intentionally extending career-related-transition-focused interventions to the 2nd year can add to the number of students who persist to the end, and graduate.

This approach is especially crucial for students with LD because of the high dropout rate reported for this group (Cortiella & Horowitz, 2014).

Implications for Practice and Future Studies

The present findings suggest that sophomore students with LD demonstrate more dysfunctional career thoughts than seniors, highlighting the need for transition programming efforts to extend beyond the 1st year of college to address the sophomore slump. When career-focused interventions and transition programming are extended to the 2nd year, we speculate that colleges and universities will likely witness an increase in the percentage of students with LD who can overcome the sophomore slump and persist to graduation. For instance, one of the critical developmental tasks college students are expected to perform is choosing a major aligned with a future career or occupation (Dipeolu, 2011; Lucas & Berkel, 2005). However, students with LD often arrive in college unprepared to make career decisions and unable to choose a major that provides direction for their college career (Dipeolu & Cook, 2006). While most universities have a career services center, students are not required to complete career development activities as part of their postsecondary education (Hitchings et al., 2001, 2005). Requiring interventions to help address these challenges, SWD could benefit from extended transitional programming that includes career-based interventions. Since SWD tend to lack knowledge about the range of occupations available and have difficulties with skills needed to choose a major that is compatible with their interests and capabilities (Lucas & Berkel, 2005), promoting career-focused programming can positively impact these students' success and likelihood of graduating within 4 years (Clayton et al., 2019; Reardon et al., 2015).

The CIP theory. The CIP theory allows practitioners to focus on critical components of self-determination such as self-knowledge, reflective thinking, planning, and organization. These components are vital to understanding the essential characteristics that help college students with LD mature to become independent decision makers and problem solvers, a necessary skill set for avoiding the sophomore slump, have a successful transition, and persist through college. This theoretical lens of CIP is also applicable when examining barriers faced by students with LD, particularly those associated with dysfunctional career thoughts as students navigate the college environment. The findings of this study suggest that the CIP approach is a useful theoretical lens when examining negative career thoughts that create barriers for sophomores with LD as they navigate their way through college. Our findings could be used when crafting transition-related programming to improve students' persistence beyond the 1st year. Career theory provides practitioners with a systematic basis for designing career interventions (Sampson, 2017). At this time, career practitioners on college campuses perceive themselves as unprepared to help address the core issue of dysfunctional career thoughts that complicate students' "career choice process or deal with the unique challenge they face (Sampson, 2017)." The successful use of the CIP theory lens in this study means practitioners can utilize the theory to understand students career needs, and use CIP-based interventions to help students identify and reframe their dysfunctional thoughts (Carr, 2004; Dieringer et al., 2017).

Limitations. The current findings should be interpreted in light of several limitations, including the use of correlational research design. This should be addressed by using research designs with more sophisticated analyses. This can include a qualitative methodology to provide a descriptive narrative of the sophomore slump and the transition of college students with LDs, particularly in the 2nd year. Furthermore, a longitudinal study that sheds more light on the unique developmental process associated with the students' transition journey, issues of dysfunctional career thoughts among students with LDs, and how persistence beyond the 1st year enhances college transitions is warranted. We agree that student retention in college in the 1st year is essential. However, future efforts to promote student success

should move from a single focus on the 1st year to embrace the process of student persistence to graduation (Tinto, 2006) and help address the sophomore slump. This topic is undoubtedly a critical research area for those scholars interested in exploring career-specific factors in understanding students' persistence in higher education. There is a need for studies such as this one to continue examining career-related gains, which is increasingly becoming more relevant to understand college persistence for students with LDs (Dong, 2019). Further study is necessary to help identify specific evidence-based intentional interventions targeting the sophomore slump.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

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Stephanie Hargrave, PhD, completed her graduate work at the Clinical Community Psychology Program, Wichita State University. She is currently the director, Institutional Research and Effectiveness and the chair of Institutional Review Board, Guilford College. Her research focus includes educational and school counseling, issues of learning differences, and disabilities. She applies these to the professional development of university faculty to help them understand student differences in order to advance the goal of student success and retention. In her spare time, she enjoys reading and working with service projects in her community.

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Yajaira A. Cabrera Tineo is currently a doctoral candidate in counseling psychology and presidential fellow in health disparities at the University at Albany, State University of New York. Her research, clinical, and teaching interests include race-based trauma, health disparities, multiculturalism, and marginalized populations. She enjoys traveling, exploring different cultures, cooking, and arts and crafts.

Ashley Longoria is a graduate student in the Psychology Program at Texas A&M University. She is currently working toward obtaining her license in counseling. She participates in Dr. Dipeolu's research team as a contributing author. She enjoys drawing, playing video games, and spending time with close family and friends outside of school and work.

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